

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CL)

November 2013

CL 315 Basic Transportation Systems

Date: 18.11.2013, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1** (a) Write in brief role of transportation in national development. [03]
(b) Write two characteristics of MAGLEV train. [02]
(c) Write four important types of signals. [02]
- Q - 2** (a) Explain preparation of sub-grade for railway track. [05]
(b) Write types and uses of dog spike and chair. [04]
(c) Enlist types of crossing. [02]
(d) Why slips are provided? Sketch double slip. [03]

OR

- Q - 2** (a) Sketch double line railway track cross section on embankment with labeling. [05]
(b) Discuss need of uniform gauge. [04]
(c) Explain gathering line with sketch. [05]
- Q - 3** (a) On a broad gauge section, a curve of 600 m radius has a limited transition of 40 m length. Calculate the maximum permissible speed and superelevation. The max. Sectional speed is 100 kmph. [05]
(b) What are the types of gradients on railways? Write objectives to provide gradients. [04]
(c) Write about terminal station with sketch. [03]
(d) Write need of interlocking. [02]

OR

- Q - 3** (a) Where and why are transition curves used in railway track? Discuss the different geometric forms adopted for transition. [05]
(b) i. What is negative superelevation? [02]
ii. Write the formula for widening of gauge on curve. [02]
(c) i. Write use of locomotive and sick yard. [05]

- ii. Write use of repeater and shunt signal.

SECTION – II

Q - 4 (a) Give classification of navigational aids. [03]

(b) Attempt any Four: [04]

- (i) Height of break water is _____ m.
- (ii) Anchoring facilities at graving dry dock, known as _____.
- (iii) Construction in form of closely spaced piles, known as _____.
- (iv) Length of graving dry dock is _____ m.
- (v) Weight of concrete block in breakwater (Rubble mound breakwater supplemented by concrete block) is _____ tones.

Q - 5 (a) Enlist site selection criteria for harbor. [04]

(b) Write functions of AAL. [05]

(c) Give classification of airway aids for traffic control. [05]

OR

Q - 5 (a) Enlist assumptions made in tide generation and explain its phenomenon. [04]

(b) Define Maximum structural landing weight and explain minimum turning radius with neat sketch. [05]

(c) Write site selection criteria for new airport and explain one. [05]

Q - 6

(a) Calculate the radius required for taxiway for a subsonic jet aircraft to negotiate the curve at a turning speed of 16666.67 mm/sec. the wheel base is 3×10^4 mm and wheel tread is 7200 mm. consider type B as per ICAO. [04]

(b) Discuss criteria of size of aircraft which decides different component sizes of airport. [04]

OR

(b) Write needs of air traffic control aids. [04]

(c) The length of runway at sea level under standard condition at zero gradient is 1500 m. The airport site is at elevation of 900 m, the airport reference temperature is 20°C. Design longitudinal profile of runway. [06]
